

Test Case Suite Report

1. Introduction

This test case suite was created to verify the correctness and reliability of the backend modules in the Rental Property Management Software. The tests focus on important business functions such as user authentication, room management, contract creation, invoice calculation, meter reading retrieval, and audit logging.

The test suite uses Jest as the testing framework and Supertest for API endpoint testing. A separate SQLite test database is used during test execution so testing does not affect real application data.

2. Testing Objectives

The main objectives of the test suite are:

- Verify that users can register, log in, refresh tokens, log out, and revoke refresh tokens correctly.
- Ensure role-based access control works properly for landlords and tenants.
- Validate room management functions such as adding, updating, viewing, and deleting rooms.
- Check contract creation rules, including room availability and tenant validity.
- Verify invoice calculation logic, including rent, electricity, water, and service fees.
- Ensure invalid billing inputs are rejected correctly.
- Test meter reading retrieval for previous billing periods.
- Verify audit log creation, filtering, and error handling.

3. Testing Tools and Environment

The backend test suite uses the following tools:

Tool	Purpose
Jest	JavaScript testing framework
Supertest	API endpoint testing
SQLite	Test database

Tool	Purpose
Node.js	Backend runtime environment

The tests are located in:

- backend/__tests__/api.test.js
- backend/__tests__/services.test.js

The test command used was:

```
npm.cmd test
```

The latest test result was:

```
Test Suites: 2 passed, 2 total
Tests:       49 passed, 49 total
```

Coverage reporting is also available through:

```
npm.cmd run test:coverage
```

The latest coverage summary was:

```
Statements   : 44.98%
Branches     : 34.60%
Functions    : 42.34%
Lines        : 46.32%
```

4. Test Scope

The test suite covers backend functionality only. Frontend UI testing is not included because the current frontend project does not have a frontend testing framework configured.

The report-generation routes are mocked during testing because PDF and Excel generation packages are not installed in the local test environment. This allows unrelated API and service tests to run reliably.

5. Test Case Summary

Test Case ID	Module	Description	Expected Result
TC-AUTH-01	Authentication	Register a new tenant	Tenant account is created with pending status
TC-AUTH-02	Authentication	Register a new landlord	Landlord account is created and tokens are returned
TC-AUTH-03	Authentication	Normalize email before saving	Email is saved in lowercase without surrounding spaces
TC-AUTH-04	Authentication	Reject duplicate email	API returns conflict error
TC-AUTH-05	Authentication	Reject missing fields	API returns validation error
TC-AUTH-06	Authentication	Reject weak password	API returns validation error
TC-AUTH-07	Authentication	Reject invalid email format	API returns validation error
TC-AUTH-08	Authentication	Login with valid credentials	Access token and refresh token are returned
TC-AUTH-09	Authentication	Login with normalized email	User can log in with uppercase or spaced email input
TC-AUTH-10	Authentication	Reject invalid login	API returns unauthorized response
TC-AUTH-11	Authentication	Refresh token rotation	New tokens are issued and old refresh token is revoked
TC-AUTH-12	Authentication	Logout user	Access and refresh tokens become invalid
TC-AUTH-13	Authentication	Revoke refresh token	Revoked refresh token can no longer be used
TC-ROOM-01	Room Management	Landlord adds a room	Room is created successfully

Test Case ID	Module	Description	Expected Result
TC-ROOM-02	Room Management	Tenant tries to add a room	Access is denied
TC-ROOM-03	Room Management	Reject room without name	API returns validation error
TC-ROOM-04	Room Management	Reject unauthenticated room creation	API returns authentication error
TC-ROOM-05	Room Management	Landlord views rooms	Room list is returned
TC-ROOM-06	Room Management	Tenant views landlord rooms	Access is denied
TC-ROOM-07	Room Management	Landlord updates room	Room details are updated
TC-ROOM-08	Room Management	Update non-existent room	API returns not found error
TC-ROOM-09	Room Management	View available rooms	Only available rooms are returned
TC-ROOM-10	Room Management	Exclude occupied room from available list	Occupied rooms are not returned
TC-ROOM-11	Room Management	Delete existing room	Room is deleted successfully
TC-ROOM-12	Room Management	Delete non-existent room	API returns not found error
TC-CONTRACT-01	Contract	Create contract for available room	Contract is created and room becomes occupied
TC-CONTRACT-02	Contract	Create contract for unavailable room	API returns room unavailable error

Test Case ID	Module	Description	Expected Result
TC-CONTRACT-03	Contract	Create contract with invalid tenant	API returns tenant invalid error
TC-BILLING-01	Billing API	Get previous meter reading	Previous month reading is returned
TC-BILLING-02	Billing API	Reject invalid month/year	API returns validation error
TC-BILLING-03	Billing API	Calculate invoice total	Correct total amount is calculated
TC-BILLING-04	Billing API	Current index lower than previous	Billing validation error is returned
TC-BILLING-05	Billing API	Missing service unit price	Service price configuration error is returned
TC-SERVICE-01	Invoice Service	Calculate invoice from valid input	Correct breakdown and total are returned
TC-SERVICE-02	Invoice Service	Missing service fees	Fees default to zero
TC-SERVICE-03	Invoice Service	Invalid numeric values	Billing validation error is thrown
TC-SERVICE-04	Invoice Service	Decreasing utility index	Billing validation error is thrown
TC-SERVICE-05	Invoice Service	Missing unit prices	Service pricing configuration error is thrown
TC-METER-01	Meter Reading Service	Retrieve latest previous reading	Correct reading is returned
TC-METER-02	Meter Reading Service	No previous reading exists	Null is returned
TC-METER-03	Meter Reading Service	Invalid room or period input	Error is thrown and database is not queried

Test Case ID	Module	Description	Expected Result
TC-AUDIT-01	Audit Log Service	Write audit event	Audit data is serialized and saved
TC-AUDIT-02	Audit Log Service	Audit insert fails	Error is handled without breaking main flow
TC-AUDIT-03	Audit Log Service	Filter audit logs	Correct filtered logs are returned
TC-AUDIT-04	Audit Log Service	Return filter options	Distinct action and entity types are returned

6. Example Test Case Description

Test Case ID: TC-BILLING-03

Module: Billing API

Test Objective: Verify that the system correctly calculates invoice totals using previous meter readings from the database.

Input Data:

Item	Value
Room price	2,500,000 VND
Previous electricity index	120
Current electricity index	150
Previous water index	60
Current water index	70
Electricity unit price	4,000 VND
Water unit price	15,000 VND
WiFi fee	100,000 VND
Trash fee	50,000 VND

Expected Result:

Item	Expected Value
Electricity usage	30 units
Water usage	10 units
Electricity fee	120,000 VND
Water fee	150,000 VND
Service fee	150,000 VND
Total invoice amount	2,920,000 VND

Actual Result: Passed.

7. Test Execution Result

After running the test suite, all test cases passed successfully.

```
PASS backend/__tests__/api.test.js
PASS backend/__tests__/services.test.js
```

```
Test Suites: 2 passed, 2 total
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```

This result shows that the tested backend functions behave as expected under both normal and invalid input conditions.

8. Resolved Limitations

The following limitation was resolved during test-suite preparation:

- Coverage reporting was added through the `test:coverage` npm script. This allows the team to measure backend test coverage without adding new dependencies.
- The API test database path was changed to a process-specific temporary SQLite file. This prevents separate test commands from locking or contaminating the same `test.db` file.

9. Remaining Limitations

The current test suite still has the following limitations:

- Frontend components are not tested because the frontend project does not currently include a UI testing framework.
- Report export functions are mocked because PDF and Excel packages are not installed in the local test environment.
- Some database integration tests still depend on SQLite behavior and are not full end-to-end tests against a production database.

These limitations do not block the current backend test suite, but they can be addressed in future iterations.

10. Conclusion

The test case suite provides automated verification for key backend features of the Rental Property Management Software. It confirms that authentication, authorization, room management, contract creation, invoice calculation, meter reading retrieval, and audit logging work correctly.

The successful execution of all 49 tests increases confidence in the stability and correctness of the backend system. The added coverage command also gives the development team a practical way to measure test coverage as the system grows.